Course notes

Throughout this chapter, coordinates are given in an orthonormal coordinate system of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \quad (t \in \mathbb{R}).$$

A point lies on the line if there exists one real t satisfying all three equalities. A line has many representations: changing the starting point or multiplying the direction vector by a non-zero real number.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Plane equations

Equation of a plane

If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a, b, c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, so it belongs to the plane.

Example. The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x - 1) - (y - 2) + 3z = 0$, i.e. $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single point of intersection; an identity gives a line contained in the plane; a contradiction gives a disjoint parallel line. For two planes, non-collinear normal vectors give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normal vectors are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use parameters s, t (independent) and solve the three equations. With no solution and non-collinear direction vectors, the lines are non-coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, find the projection of M on the normal line through M : $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learnt in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Checking is essential. After calculating a projection, check simultaneously $H \in \mathcal{P}$ and the collinearity of $\overrightarrow{MH}$ with the normal vector.

Moving from objects to equations

12.1 One line, several parametrisations

First use: S65.

Definition · Parametrising a line

A line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$, with $t \in \mathbb{R}$. The parameter is neither a prescribed coordinate nor a time, unless the context specifies it.

Worked example · Same set, other parameters

The expressions $(1 + t; 2 - t; 2t)$ and $(3 + 2s; -2s; 4 + 4s)$ describe the same line: set $t = 2 + 2s$ transforms the first into the second, and this transformation runs through all real numbers. The starting points may differ, and the direction vectors may be proportional.

A1 · Immediate application Show that $(t; 1 + 2t; 3 - t)$ and $(2 + 3s; 5 + 6s; 1 - 3s)$ describe the same line. Is the point $(1; 3; 2)$ belong to it?

12.2 A normal vector fixes the equation of a plane

First use: S67.

Property · Cartesian equation

In an orthonormal coordinate system, a plane with nonzero normal vector $(a; b; c)$ through $A = (x_A; y_A; z_A)$ has equation $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. It expresses $\overrightarrow{AM} \cdot \vec{n} = 0$.

Worked example · Construct, then test

For A=(1;0;2) and $\vec{n} = (2; -1; 3)$, the plane has equation 2 $x - y + 3z - 8 = 0$. The point (0;1;3) lies in the plane since 0 $-1 + 9 - 8 = 0$. A point does not lie in the plane just because its triple resembles the normal vector.

A2 · Partial guidance Determine the plane with normal vector $(1; 2; -1)$ through (2;0;1). Test whether (1;1;2) lies in it, then give two non-collinear directions of the plane.

12.3 Three outcomes for a single equation

First use: S69.

After substitution	Interpretation	Example
$at + b = 0$, $a \neq 0$	a point	intersecting line
$0t + b = 0$, $b \neq 0$	no point	strictly parallel
$0t + 0 = 0$	all points	line in the plane

Worked example · Reading the equation y = mx + p

For the plane x+y+z=1 and the line $(t;2t;1-t)$, substitution gives 2 $t+1=1$, so t=0 and a single intersection (0;0;1). We conclude only after substituting the value back into all three coordinates.

A3 · Compare In the plane $z=1$, study the relative positions of the lines $d_1 : (t;0;t)$, $d_2 : (t;0;0)$ and $d_3 : (t;0;1)$. Give the intersection point when it is unique and sketch the three situations.

12.4 Two lines, two independent parameters

First use: S70.

Worked example · Do not impose a false simultaneity

For $d : (t;0;0)$ and $d' : (1;1+s;0)$, the intersection requires t=1 and s=-1. Imposing t=s would wrongly eliminate the point (1;0;0). The parameters are used separately to run along each line.

A4 · Fix a method For $d : (t;t;0)$ and $d' : (1+s;1-s;s)$, solve with two parameters, then explain why all three coordinates must be checked.

12.5 Two checks after the system

First use: S70.

Worked example · Projecting onto a plane

Let $\mathcal{P} : x+2y-z=1$ and M=(1;1;1). The normal line through M is $H=(1+t;1+2t;1-t)$. Substituting into the plane gives 2 $+6t=1$, therefore $t=-1/6$. H=(5/6;2/3;7/6). We check 5 $/6+4/3-7/6=1$ and $\overrightarrow{MH}=(-1/6)(1;2;-1)$. The distance is $\overline{6}/6$.

A5 · Synthesis with a degenerate case For a real number m, d_m passes through (0;0;m) and has direction $(1;m;1)$. Study the intersection with $x+2y-z=1$. Treat m=0 separately. Is there an m for which the line is perpendicular to the plane?

A first step: Substitute the three coordinates before dividing. For orthogonality, compare the direction vector with a normal vector.

A6 · Choice of strategy Determine the intersection of the planes $x+y+z=3$ and $x-y=0$. Give a point and a direction vector, then check these two pieces of information in the equations.

One minute, spoken aloud. State the two properties that make it possible to recognise a projection without mentioning coordinates.

Delayed review · S71 then S87. Match each line of an algebraic calculation with a geometric property; deal first with the parameters that make a denominator zero.

Investigation

Three lines and a plane - 30 min.

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x+y+z=3$. In each question, the parameter t runs through $\mathbb{R}$.

1. The line d_1 has the representation $(x;y;z)=(t;t;t)$. Determine $d_1 \cap \mathcal{P}$.

2. The line d_2 has the representation $(x;y;z) = (3+t;-t;0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line d_3 has the representation $(x;y;z) = (t;-t;0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, state whether the equation obtained after substituting into the plane's equation has a unique solution, no solution or every real number as a solution. Relate these cases to the geometric positions found.